

Learning DiaryGrocery Application

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1 Project Overview

The purpose of this project was to learn Android application development using Java and Android Studio. The application allows users to add grocery items, mark them as important, and display all items in a list.

2 What I Learned

During this project I learned how Android applications are structured and how different components communicate with each other.

I learned how to:

- Create and use Fragments.
- Use ViewPager2 and TabLayout for navigation.
- Create user interfaces using XML layouts.
- Handle button click events.
- Read data from EditText fields.
- Use CheckBox components.
- Display data using RecyclerView.
- Create custom adapters for RecyclerView.
- Store and manage objects using Java classes.

3 Challenges

One challenge was understanding how Fragments work together with ViewPager2. Initially, some views overlapped with the TabLayout because of layout configuration issues.

Another challenge was updating the RecyclerView correctly after adding new grocery items.

I also spent time learning how adapters and ViewHolders are used to display data efficiently.

4 Improvements

I added functionality that automatically clears the input fields after a grocery item is added. I also added feedback messages using Toast notifications.

If I continue developing the project, I would like to:

- Save grocery items permanently using a database.
- Add the ability to delete items.
- Add item editing functionality.
- Improve the visual design and styling.

5 Conclusion

This project improved my understanding of Android application development and object-oriented programming. I gained practical experience with user interface design, navigation between views, and displaying dynamic data using RecyclerView.